Fashion-MNIST Image Compression and Reconstruction Using Autoencoders

Deep Learning (2087101) — Concise Project Report

1. Aim

To implement an undercomplete autoencoder for Fashion-MNIST image compression, compare latent dimensions 8, 32, and 64, and implement a denoising autoencoder for noisy images.

2. Objectives

• Build an encoder-decoder network with a bottleneck smaller than the input.
• Measure test-set Mean Squared Error (MSE).
• Compare compression and reconstruction for three latent sizes.
• Train a denoising autoencoder using noisy inputs and clean target images.

3. Dataset and preprocessing

Fashion-MNIST contains 28 × 28 grayscale clothing images. Each image has 784 pixel values. Pixel values are normalized from 0–255 to 0–1 and flattened into a vector of length 784. Labels are not needed because the autoencoder learns without labels.

4. Architecture

Encoder: 784 input values → Dense(128, ReLU) → latent layer (8, 32, or 64 units).

Decoder: latent vector → Dense(128, ReLU) → Dense(784, sigmoid).

The model is trained to reproduce its input using Adam and mean squared error loss. For denoising, noisy images are inputs and the original clean images are targets.

5. Compression ratio

Theoretical value-count compression ratio = input dimension / latent dimension = 784 / latent size. Therefore, latent size 8 gives 98:1, size 32 gives 24.5:1, and size 64 gives 12.25:1. These ratios compare input pixel values with latent values only; they do not include model weights, metadata, or the bit width of values.

6. Evaluation

The notebook calculates test MSE as the mean of (original pixel − reconstructed pixel)² across all test pixels. It also plots training/validation loss and visualizes original and reconstructed images. The denoising section compares original, noisy, and denoised images.

7. Results table

Run all cells in the notebook and enter the measured MSE values below. Values are intentionally left blank rather than inventing experimental results.

Latent size	Value-count ratio
8	98:1
32	24.5:1
64	12.25:1
Denoising (32)	24.5:1*

*For denoising, MSE is measured against the clean test images. The listed ratio is the input-value count ratio only.

8. Analysis and observations

A smaller bottleneck stores fewer values and provides stronger compression, but may discard details needed for accurate reconstruction. A larger bottleneck can preserve more image information and often lowers MSE, although the measured result depends on training settings. The denoising autoencoder learns to reconstruct clean images from corrupted inputs.

9. Conclusion

The project demonstrates unsupervised image compression and reconstruction with an undercomplete autoencoder and noise removal with a denoising autoencoder. The latent-size comparison illustrates the trade-off between compact representations and reconstruction quality. Final MSE values should be recorded from the executed notebook.

10. Files and execution

The accompanying notebook includes the model code, loss curves, test-set evaluation, and image visualizations. It can be run in Google Colab or Jupyter with TensorFlow, NumPy, and Matplotlib installed.